Photograph the Moon

Print it, fold it, take it out with you.

Fragments of the Universe
bhapstar.com

The whole technique, in one paragraph

The Moon is a sunlit rock, so expose it like something in daylight, not like the night sky. Set the camera to manual, f/8 to f/11, ISO 100 to 200, and a shutter of one over the ISO number. Nothing needs to track, because at those speeds the sky has not moved. Then pick a night that is not full: all the detail lives along the terminator, the curved line between the lit and unlit halves, where the shadows are long.

IS IT UP TONIGHT, AND IS TONIGHT WORTH IT

Crescent	First qtr	Gibbous	Full	Gibbous	Last qtr	Crescent	New
day 3	day 7	day 11	day 15	day 18	day 22	day 26	day 0
until 9pm	until midnight	until 3am	all night	from 9pm	from midnight	from 3am	daytime only
plenty of relief	shadows at their longest	plenty of relief	no shadows facing you	plenty of relief	shadows at their longest	plenty of relief	go shoot the Milky Way

The Moon rises about fifty minutes later each day, so it works its way round the clock once a month. Times above are nominal and move with your latitude and the season, but the order never changes: the first half of the month is an evening Moon, the second half is a small-hours Moon, and for three or four nights around new there is nothing to photograph at all. Those are the nights the rest of astrophotography waits for.

CAMERA ON A TRIPOD, ALL MANUAL

Mode	Manual	Every auto mode meters the black sky and blows the Moon out
Aperture	f/8 to f/11	Most lenses are sharpest a stop or two down from wide open
ISO	100 to 200	There is plenty of light. Low ISO keeps the file clean
Shutter	One over the ISO	ISO 200 means 1/200s. The Looney 11 rule. Then adjust by eye
Focus	Manual, magnified	Put a crater edge or the terminator centre screen and turn until the shadows snap. Re-check after half an hour as the lens cools
File	RAW	Keeps the detail in the bright areas that a JPEG throws away
Drive	Burst of 20 or more	The air is moving. Some frames caught it holding still
Stabilisation	Off	On a tripod it can add the shake it exists to remove
Tracking	Not needed	At 1/200s the sky has not moved. A wall will do

A crescent wants a little more light than a full Moon, because you are seeing the surface at a low angle. Start at the numbers above and open up half a stop at a time until the terminator stops looking empty.

WITHOUT A CAMERA

Phone on its own

Tap the Moon and hold until the exposure locks, then drag the slider a long way down until grey markings appear. Use the telephoto lens if the phone has one, never the digital zoom.

Phone at an eyepiece

The cheapest big jump there is. Focus by eye first, then hold the main camera lens square to the eyepiece and close in. A dark ring means you are too far back. Lock the exposure, drag it down, shoot a burst.

Smart telescope

Use the mode meant for bright targets. It drops the exposure right down and tracks while you watch. Aperture matters far less on the Moon than on a faint galaxy, so a small one does well here.

HOW MUCH OF THE FRAME THE MOON FILLS

100mm	200mm	400mm	800mm	1600mm	2700mm
4%	8%	15%	30%	60%	100%
a dot	a small disc	croppable	a real picture	craters everywhere	edge to edge

Measured on a full-frame sensor. A crop body shows the Moon about half again as large at the same focal length, and a small astronomy camera larger still, because both are cropping into the middle of the same picture.

WHAT LIMITS YOU IS THE AIR

Wait until it is high

Near the horizon you are looking through several times more air. Thirty degrees up is a sensible minimum. Higher is better.

Never over a hot roof

A road, a rooftop or an air conditioning unit that baked all day pours heat upwards all evening. In a city this ruins more nights than light pollution does.

Stack the sharpest

Shoot a burst or a short video, throw away the soft frames and average the rest. Siril does it free. That is how sharp lunar pictures are made.

BEFORE YOU PRESS ANYTHING

- **Phase** not full
- **Altitude** over 30 degrees
- **Exposure** manual, way down
- **Focus** magnified, by hand
- **Frames** a burst, not one

The full guide, with examples

Full walkthrough with example images, the phase timetable in detail, how to tell in ten seconds whether the Moon is up tonight, and what to photograph on the few nights each month when it is not.

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